

UNDERSTANDING DEEP LEARNING REQUIRES RE-THINKING GENERALIZATION

Chiyuan Zhang*

Massachusetts Institute of Technology
chiyuan@mit.edu

Samy Bengio

Google Brain
bengio@google.com

Moritz Hardt

Google Brain
mrtz@google.com

Benjamin Recht†

University of California, Berkeley
brecht@berkeley.edu

Oriol Vinyals

Google DeepMind
vinyals@google.com

ABSTRACT

Despite their massive size, successful deep artificial neural networks can exhibit a remarkably small difference between training and test performance. Conventional wisdom attributes small generalization error either to properties of the model family, or to the regularization techniques used during training.

Through extensive systematic experiments, we show how these traditional approaches fail to explain why large neural networks generalize well in practice. Specifically, our experiments establish that state-of-the-art convolutional networks for image classification trained with stochastic gradient methods easily fit a random labeling of the training data. This phenomenon is qualitatively unaffected by explicit regularization, and occurs even if we replace the true images by completely unstructured random noise. We corroborate these experimental findings with a theoretical construction showing that simple depth two neural networks already have perfect finite sample expressivity as soon as the number of parameters exceeds the number of data points as it usually does in practice.

We interpret our experimental findings by comparison with traditional models.

1 INTRODUCTION

Deep artificial neural networks often have far more trainable model parameters than the number of samples they are trained on. Nonetheless, some of these models exhibit remarkably small *generalization error*, i.e., difference between “training error” and “test error”. At the same time, it is certainly easy to come up with natural model architectures that generalize poorly. What is it then that distinguishes neural networks that generalize well from those that don’t? A satisfying answer to this question would not only help to make neural networks more interpretable, but it might also lead to more principled and reliable model architecture design.

To answer such a question, statistical learning theory has proposed a number of different complexity measures that are capable of controlling generalization error. These include VC dimension (Vapnik, 1998), Rademacher complexity (Bartlett & Mendelson, 2003), and uniform stability (Mukherjee et al., 2002; Bousquet & Elisseeff, 2002; Poggio et al., 2004). Moreover, when the number of parameters is large, theory suggests that some form of regularization is needed to ensure small generalization error. Regularization may also be implicit as is the case with early stopping.

1.1 OUR CONTRIBUTIONS

In this work, we problematize the traditional view of generalization by showing that it is incapable of distinguishing between different neural networks that have radically different generalization performance.

*Work performed while interning at Google Brain.

†Work performed at Google Brain.